

ARCHITECTURE_Latency

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Definition

In an imaging or vision system

- What is the delay incurred, between the start of the capture process to the completion of the display result?

3 Stages

- Camera Time - Begin: Give command to capture; End: Start network transmission of data from router
- Channel Time - Begin: Start network transmission of data; End: Jitter = All packets reassembled as a frame
- Display Time - Begin: Packets reassembled as frame; End: Last row of frame displayed on screen

Camera

For this example, assume a 30 fps video stream.

- 1) Give command to capture to actual start of capture: 0 to 33 msec, depending on timing with sync
- 2) Exposure time: Worse case - $1/30 = 33$ msec; Better - $1/100 = 10$ msec
- 3) Rolling shutter readout time: 10 msec, typical
- 4) MIPI transmission: 2 msec, typical

Camera

- ISP: 8 msec, typical
- GPU: Buffering: 33 msec - 66 msec, typical
- Encoder: 20 msec, typical

Channel

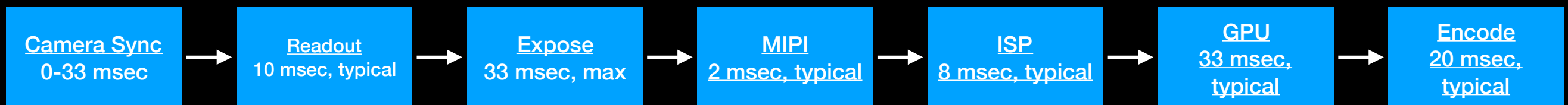
- Baseline one way network delay (US): 25-60 msec
- Observed delay variation (jitter, WiFi): 20-50 msec
- Best case: Ethernet; Worse Case: Cellular
- Note: Channel latency can dominate Camera and Display latency - Example: Intercontinental & Cellular = 150 msec + 150 msec

Display

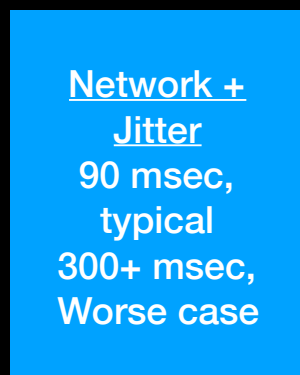
- Decoder: 10 msec, typical
- Display processing: 10 msec (game mode), typical; 30 - 50 msec (non-game mode), typical
- Display sync (60 Hz display): 0 - 16 msec, 8 msec typical
- Display draw (60 Hz display): 16 msec

Summary

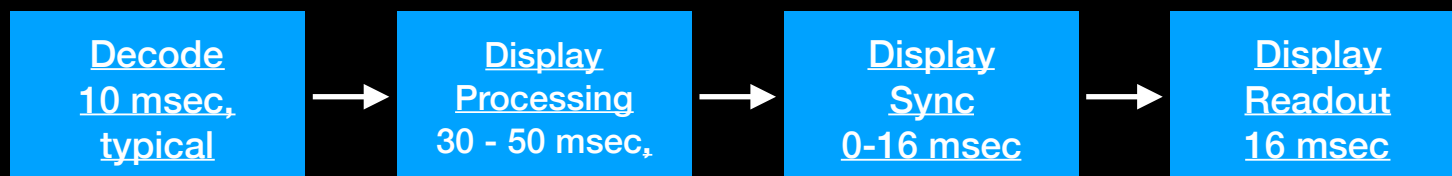
Camera = 16 + 10 + 10 + 2 + 8 + 33 + 20 = 99 msec, typical



Channel = 90 msec, typical



Display = 10 + 40 + 8 + 16 = 64 msec, typical



Measurement Setup

